

Finite Automata Sheet

Gallo Tennis

April 19, 2025

Automata theory problems from the book "Automata Theory, Languages, and Computation" by J. Ullman.

NFA

1. **(Equivalence theorem.)** If $D = (Q_D, \Sigma, \delta_D, \{q_0\}, F_D)$ is the DFA constructed from NFA $N = (Q_N, \Sigma, \delta_N, q_0, F_N)$ by the subset construction, then $L(D) = L(N)$. Conversely if L is a language that is accepted by some DFA then it is accepted by some NFA.

Proof. By definition of language of both N and a generic DFA

$$L(N) := \{w \in \Sigma^* : \hat{\delta}_N(q_0, w) \cap F_N \neq \emptyset\}$$

$$L(D) := \{w \in \Sigma^* : \hat{\delta}_D(\{q_0\}, w) \in F_D\}$$

In our case, let $S = \hat{\delta}_D(\{q_0\}, w)$ (which is a state made of a set of F_N), then

$$S \in F_D \iff S \cap F_N \neq \emptyset.$$

In other words, the only thing to prove is that for any $w \in \Sigma^n$

$$\hat{\delta}_D(\{q_0\}, w) = \hat{\delta}_N(q_0, w).$$

We need to rely in the transition function constructed by subsets out of $\hat{\delta}_N$. By induction on $n = |w|$:

Basis. Assume $n = 0$ so that $w = \varepsilon$. Then the construction by subsets transition function will simply be $\delta_D(\{q_0\}, \varepsilon) = \{q_0\}$ then also we have

$$\hat{\delta}_D(\{q_0\}, \varepsilon) = q_0 = \hat{\delta}_N(q_0, \varepsilon).$$

Thus we conclude that $\delta_D = \hat{\delta}_D = \hat{\delta}_N$ (under renaming the singleton set $\{q_0\}$ as q_0).

Hypothesis. Assume that for the string x of length n it held that

$$\hat{\delta}_D(\{q_0\}, x) = \hat{\delta}_N(q_0, x) = \{p_1, \dots, p_m\} = S.$$

Thesis. We have to prove that for a string $w = xa$ such that $n = |x|$ it follows that

$$\hat{\delta}_D(\{q_0\}, w) = \hat{\delta}_N(q_0, w).$$

To prove this assertion we recall the definition of $\hat{\delta}_D$ for any DFA and compare it to the one constructed by subsets. We have that

$$\hat{\delta}_D(\{q_0\}, w) \stackrel{def}{=} \delta_D(\hat{\delta}_D(\{q_0\}, x), a) \quad (1)$$

$$\delta_D(S, a) \stackrel{c.b.s.}{=} \bigcup_{p \in S} \delta_N(p, a) = \bigcup_{i=1}^m \delta_N(p_i, a). \quad (2)$$

Note that in (2), by construction S is a subset of Q_N such that

$$\hat{\delta}_D(\{q_0\}, w) = \delta_D(\hat{\delta}_D(\{q_0\}, x), a) = \delta_D(S, a) \quad (3)$$

hence, the transition function constructed by subsets is a valid transition function for D . Furthermore by equation (1)

$$\hat{\delta}_D(\{q_0\}, w) = \bigcup_{i=1}^m \delta_N(p_i, a).$$

We also note that in the case of the extended transition function of N

$$\hat{\delta}_N(q_0, w) \stackrel{def}{=} \bigcup_{i=1}^m \delta_N(p_i, a) = \delta_D(S, a) \stackrel{(3)}{=} \hat{\delta}_D(\{q_0\}, w).$$

Conversely, to prove that a DFA $D = (Q_D, \Sigma, \delta_D, \{q_0\}, F_D)$ induced language is the same as some NFA language, by structural induction on n , the length of the input string, we define the NFA N as

Basis. We let the start state q_0 of D be the start state also of N . So that for $w = \varepsilon$, $\hat{\delta}_D(\{q_0\}, \varepsilon) = \hat{\delta}_N(q_0, \varepsilon)$.

Hypothesis. For $|x| = n$ suppose $\hat{\delta}_D(\{q_0\}, x) = p_n$ then we let $\hat{\delta}_N(q_0, x) = \{p_n\}$.

Thesis. For $w = xa$ with $|w| = n + 1$, by hypothesis

$$\hat{\delta}_D(\{q_0\}, w) = \delta_D(\hat{\delta}_D(\{q_0\}, x), a) = \delta_D(p_n, a) = p_{n+1}.$$

We let $\hat{\delta}_N(q_0, w) = \{p_{n+1}\}$.

□

2. **(Worst case.)** Show that there exists a NFA with n states whose equivalent DFA has exactly 2^n states.

Solution. We shall build an NFA that accepts univocally the language

$$\{w \in \{0, 1\}^* : \text{the } n\text{-th symbol from the end of } w \text{ is } 1\}.$$

We first construct an NFA of this language having n states. By structural induction on $|w|$

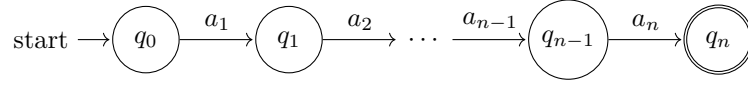
Basis.

Hypothesis.

Thesis.

3. Prove that if N is an NFA that has at most one choice of state for any state and input symbol (i.e., $\delta(q, a)$ never has size greater than 1), then the DFA D constructed from N by the subset construction has exactly the states and transitions of N plus transitions to a new dead state whenever N is missing a transition for a given state and input symbol.

Solution. Let $Q_N = q_0, q_1, \dots, q_n$. Consider the maximal case in which each state except possibly the last one has exactly one transition. Therefore the automaton is isomorphic to



It is clear that the NFA described must accept only strings w of length n . Else, for the sake of contradiction, suppose that the length were $k > n$ then, by pigeonhole principle there must exist at least one state q_i such that for a symbol a_i ; $\delta_N(q_i, a_i) = \{p, r\}$, contradicting the fact that $\delta_N(q, a)$ is a singleton (supposing that $k < n$ yields the same contradiction, but we get an unused state instead).

We claim that $\hat{\delta}_N(q_0, w) = \{q_n\}$ this is, $\hat{\delta}_N(q_0, w)$ is a singleton. To prove this claim, by induction on n the amount of states (not counting the starting one) and the length of the accepted strings

Basis. If $n = 1$, then by problem hypothesis

$$\hat{\delta}_N(q_0, a_1) = \delta_N(q_0, a_1) = \{q_1\}.$$

Hypothesis. Assume that for n states and for a string $x = a_1 \dots a_n$ we had

$$\hat{\delta}_N(q_0, x) = \{q_n\}.$$

Thesis. We have to prove that for $n + 1$ states and for a string xa_{n+1}

$$\hat{\delta}_N(q_0, xa_{n+1}) = \{q_{n+1}\}.$$

By hypothesis,

$$\hat{\delta}_N(q_0, xa_{n+1}) = \bigcup_{p \in \hat{\delta}_N(q_0, x)} \delta_N(p, a_{n+1}) = \bigcup_{p \in \{q_n\}} \delta_N(p, a_{n+1}) = \delta_N(q_n, a_{n+1})$$

so by construction,

$$\delta_N(q_n, a_{n+1}) = \{q_{n+1}\}.$$

By induction on k the length of any accepted string (and the amount of states distincts from the starting one), we claim that the equivalent DFA yields the same states as the NFA plus a dead state.

Basis. For $k = 2$ consider the transition table for the NFA,

	a_1	a_2
$\rightarrow q_0$	$\{q_1\}$	$\emptyset$
q_1	$\emptyset$	$\{q_2\}$
$*q_2$	$\emptyset$	$\emptyset$

Then, the transition table for the equivalent DFA will be

	a_1	a_2
$\emptyset$	$\emptyset$	$\emptyset$
$\rightarrow q_0$	$\{q_1\}$	$\emptyset$
q_1	$\emptyset$	$\{q_2\}$
$*q_2$	$\emptyset$	$\emptyset$
$\{q_0, q_1\}$	$\{q_1\}$	$\{q_2\}$
$*\{q_0, q_2\}$	$\{q_1\}$	$\emptyset$
$*\{q_1, q_2\}$	$\emptyset$	$\{q_2\}$
$*\{q_0, q_1, q_2\}$	$\{q_1\}$	$\{q_2\}$

Changing names of states we see that

	a_1	a_2
$\emptyset$	$\emptyset$	$\emptyset$
$\rightarrow q_0$	$\{q_1\}$	$\emptyset$
q_1	$\emptyset$	$\{q_2\}$
$*q_2$	$\emptyset$	$\emptyset$
A	$\{q_1\}$	$\{q_2\}$
$*B$	$\{q_1\}$	$\emptyset$
$*C$	$\emptyset$	$\{q_2\}$
$*D$	$\{q_1\}$	$\{q_2\}$

so that in particular, A and D are the same state, B is the same state as q_0 and C is the same state as q_1 . Furthermore, we see that A, D are unreachable. We conclude that for $k = 2$ the transition table for the NFA is

	a_1	a_2
$\emptyset$	$\emptyset$	$\emptyset$
$\rightarrow q_0$	$\{q_1\}$	$\emptyset$
q_1	$\emptyset$	$\{q_2\}$
$*q_2$	$\emptyset$	$\emptyset$

Hypothesis. Assume that for a string $x = a_1 \dots a_k$ of length k it held that for an NFA N with states $Q_N = \{q_0, \dots, q_k\}$ the equivalent DFA had same k states as the NFA plus one dead state: $Q_D = \{q_0, \dots, q_k, \emptyset\}$.

Thesis. Consider the string $a_1 \dots a_k a_{k+1} = x a_{k+1}$, this case can be seen by adding a column a_{k+1} , hence by the first claim,

$$\hat{\delta}_N(q_0, x a_{k+1}) = \bigcup_{i=1}^k \delta_N(q_i, a_{k+1}) = \{q_{k+1}\}$$

so that the DFA also transitions to the singleton $\{q_{k+1}\}$ only in the state q_k by subset construction. Then, necessarily q_{k+1} transitions to $\emptyset$ since there are not further states.

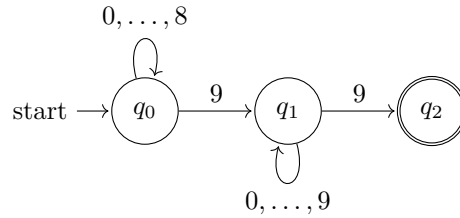
We see that by hypothesis all the subsets of $\{q_0, \dots, q_k\}$ that are not singletons are unreachable, then since the only reachable state starting from q_k is q_{k+1} and q_{k+1} transitions to $\emptyset$, it follows that any subset state of $\{q_1, \dots, q_k, q_{k+1}\}$ is unreachable.

□

4. Give nondeterministic finite automata to accept the following languages. Try to take advantage of nondeterminism as much as possible.
 - (a) The set of strings over alphabet $\{0, 1, \dots, 9\}$ such that the final digit has appeared before.
 - (b) The set of strings over alphabet $\{0, 1, \dots, 9\}$ such that the final digit has not appeared before.
 - (c) The set of strings of 0's and 1's such that there are two 0's separated by a number of positions that is a multiple of 4. Note that 0 is an allowable multiple of 4.

Solution.

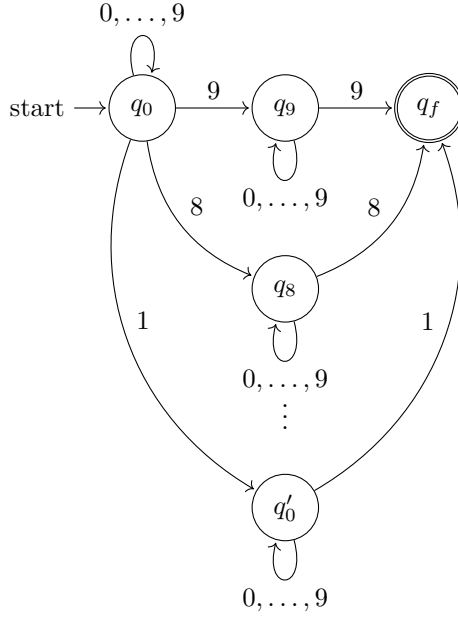
- (a) Consider the simpler automaton that accepts any string that ends in 9 and also has at least a 9 before.



The state q_0 handles all the symbols that are not 9, when it handles the symbol 9 it transitions to q_1 , this means that “in between” the string there is a 9. Then, q_1 will not transition until it “guesses” it has finished with a 9.

We can build each automaton of this form for each number from 0 to 9. The final automaton needs to start at q_0 and bifurcate in 9 branches, each of whom will be the above automaton.

Hence the automaton is given by



In this case q_0 has a loop with transitions for any $0, \dots, 9$ because it handles the cases in which we do not know the ending digit, if we did not put that loop, then it would mean that the first digit must be the last one, which is not necessarily the language we want to compute.

- (b) This NFA can be done normally or with epsilon transitions, with epsilon transitions the amount of states of this automaton will be fewer.

We let the starting state q_0 have an epsilon transition to 10 possible branches, the first state of each of these branches will be called S_i for $0 \leq i \leq 9$ and will represent the guess that the digit i is not in the prefix of the string.

Let A_i be an accepting state in each of the branches. We define our simple transition function for any symbol $a = 0, \dots, 9$ as

$$\delta(S_i, a) := \begin{cases} S_i & \text{if } a \neq i \\ A_i & \text{if } a = i \end{cases}$$

and for handling that the branches “guess” was wrong (the symbol appeared in the prefix, or did not know what the last symbol was), that branch must die

$$\delta(A_i, a) := \emptyset \text{ for any } a \in \Sigma.$$

By induction on $|w|$ we observe the behaviour of the extended transition function.

Basis. If $|w| = 1$ so that $w = a$, then

$$\begin{aligned}\hat{\delta}(q_0, \varepsilon a) &= \text{ECLOSE} \left(\bigcup_{i=0}^9 \delta(S_i, a) \right) \\ &= \text{ECLOSE}(\delta(S_0, a) \cup \dots \cup \delta(S_a, a) \cup \dots \cup \delta(S_9, a))\end{aligned}$$

which by construction results in

$$\text{ECLOSE}(S_0 \cup \dots \cup A_a \cup \dots S_9) = \{S_0, \dots, S_{a-1}, A_a, S_{a+1}, \dots, S_9\}.$$

Hypothesis. Assume that for $|w| = n$ with $w = xa$ it holds that if $a \notin x$, then

$$\hat{\delta}(q_0, xa) = \{S_0, \dots, S_{a-1}, A_a, S_{a+1}, \dots, S_9\},$$

and if $a \in x$ then $\hat{\delta}(q_0, w)$ does not contain any accepting state A_i for no i nor contains S_a (the branch became unusable).

Thesis. Suppose we have a string $w' = xab$ such that $|w'| = n + 1$. For the case $a \notin x$, by hypothesis

$$\hat{\delta}(q_0, xa) = \{S_0, \dots, S_{a-1}, A_a, S_{a+1}, \dots, S_9\}$$

therefore if $b = a$ or $b \in x$, by construction we have that

$$\delta(A_a, b) = \delta(A_a, a) = \emptyset$$

so in particular we have that

$$\hat{\delta}(q_0, xab) = \bigcup_{p \in \hat{\delta}(q_0, xa)} \delta(p, b) = \{S_0, \dots, S_{a-1}, S_{a+1}, \dots, S_9\}.$$

Else, if $b \neq a$ and $b \notin x$, A_a must die and A_b must remain:

$$\hat{\delta}(q_0, xab) = \bigcup_{p \in \hat{\delta}(q_0, xa)} \delta(p, b) = \{S_0, \dots, S_{b-1}, A_b, S_{b+1}, \dots, S_9\}.$$

The last ECLOSE yields the same results, so it is unnecessary.

For the case $a \in x$, by hypothesis

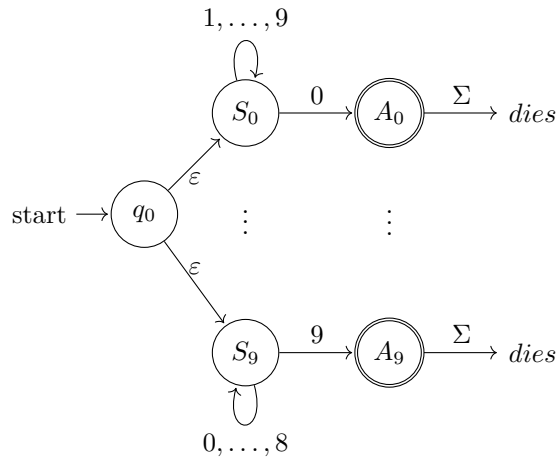
$$\hat{\delta}(q_0, xa) = \{S_0, \dots, S_{a-1}, S_{a+1}, S_9\}$$

then, if $b = a$ nothing changes. If $b \neq a$ then

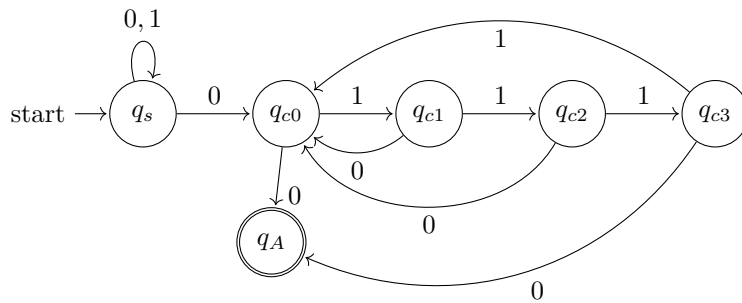
$$A_b \in \hat{\delta}(q_0, xab).$$

and there is no other $A_i \in \hat{\delta}(q_0, xab)$ for no i .

A transition diagram of this automaton is



(c) The automaton is given by the following transition diagram



□

ε -NFA

1. (**Equivalence theorem.**) A language L is accepted by some ε -NFA if and only if L is accepted by some DFA.

Solution

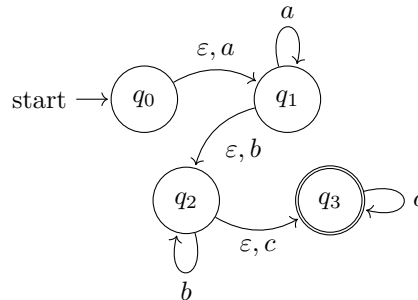
($\Rightarrow$)

2. Design ε -NFA's for the following languages. Try to use ε -transitions to simplify your design.
 - (a) The set of strings consisting of zero or more a 's followed by zero or more b 's, followed by zero or more c 's.
 - (b) The set of strings that consist of either 01 repeated one or more times or 010 repeated one or more times.

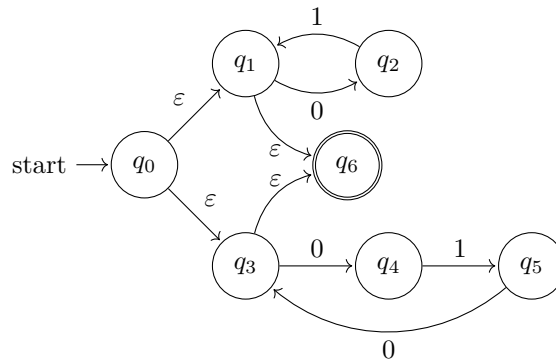
- (c) The set of strings of 0's and 1's such that at least one of the last ten positions is a 1.

Solution.

- (a) This automaton will have the following transition diagram, the design is self-explanatory



- (b) In this case we want to bifurcate in two possible paths, one of them will handle the strings of the form 0101... and the other will handle the strings of the form 010010.... We can guess what the string will be using an epsilon transition at the beginning.



- (c) We will have at least

Regular expressions

- Write regular expressions for the following languages:
 - The set of strings over alphabet $\{a, b, c\}$ containing at least one a and at least one b .
 - The set of strings of 0's and 1's whose tenth symbol from the right end is 1.

- (c) The set of strings of 0's and 1's with at most one pair of consecutive 1's.
2. Write regular expressions for the following languages:
- (a) The set of all strings of 0's and 1's such that every pair of adjacent 0's appears before any pair of adjacent 1's.
 - (b) The set of strings of 0's and 1's whose number of 0's is divisible by five.
3. Write regular expressions for the following languages:
- (a) The set of all strings of 0s and 1s not containing 101 as a substring.
 - (b) The set of all strings with an equal number of 0s and 1s, such that no prefix has two more 0s than 1s, nor two more 1s than 0s.
 - (c) The set of strings of 0s and 1s whose number of 0s is divisible by five and whose number of 1s is even.